Bhoomika Ravishankar

447-902-6867 | itsbhoomika.r@gmail.com | Austin, TX | LinkedIn | GitHub | Portfolio

Education

University of Illinois Urbana-Champaign May 2026
M.S. in Information Management; GPA: 4.0/4.0; Phi Kappa Phi Honor Society *Urbana-Champaign, IL*
- **Coursework:** Advanced NLP, Text Mining, Statistical Models, Database Design & Prototyping; recognized among the **top 10% of graduate students** and won a university AI hackathon.

Savitribai Phule Pune University Graduated with Distinction
B.E. in Computer Engineering; GPA: 9.12/10 *Pune, India*
- **Coursework:** Data Structures, Database Management, Machine Learning, Big Data Analytics, Cloud Computing.

Work Experience

Synchrony Financial May 2025 – May 2026
Data Science Intern *Champaign, IL*
- Owned a **100M+ row** customer-analytics workflow: repaired upstream data issues, imputed **100K+ missing values**, and replaced a vendor-led model with an internal XGBoost, Lasso, causal-inference, and SHAP pipeline.
- Defined success criteria and identified **five customer-experience drivers** of Net Promoter Score, converting model findings into a leadership-ready decision narrative.
- Engineered **13+ reusable macroeconomic and credit-risk features** in PySpark, published curated datasets to AWS S3, and analyzed model behavior across **900M account records** and two retail portfolios.
- Reduced validation cycle time by **25%** for a **$7.49B loss model** by backtesting 15+ years of account snapshots, surfacing failure modes, and informing rapid model refinements.

UIUC Business Intelligence Group Jan. 2025 – May 2025
Lead Data Consultant *Champaign, IL*
- Led client-facing delivery of a RAG prototype, translating ambiguous stakeholder questions into specifications, retrieval success criteria, and demo-feedback cycles that reduced manual reporting by **20%**.
- Owned architecture and quality control across ingestion, chunking, intent classification, grounded retrieval, hallucination safeguards, Streamlit delivery, stakeholder demos, and documentation.

Repute Engineers Pvt. Ltd. Feb. 2023 – Apr. 2023
Data Analytics Intern *India*
- Reconciled fragmented CSV, ERP, finance, BOM, and production data across **500+ projects**; automated **50+ SQL Server/Power BI KPIs** and presented cost recommendations to VP-level stakeholders.

Research Experience

Prof. Ruoqing Zhu Lab, UIUC Department of Statistics Jan. 2025 – Present
Research Data Scientist *Champaign, IL*
- Built a configurable Python/Bash ETL pipeline on an HPC cluster to harmonize fragmented clinical sources using schema mapping, normalization, quality checks, one-hot encoding, and PCA.
- Developed an LLM-assisted modeling loop over **115K+ cancer trials**; converted narratives into structured features whose attributes represented **97.2% of feature importance** in a Lasso logistic benchmark.

Selected Projects

Agentic Pipeline Optimizer | *LangChain, Ollama, Airflow, dbt, BigQuery, ChromaDB, Docker* GitHub
- Architected a Dockerized three-agent workflow that diagnoses pipeline failures, queries Airflow/dbt metadata over BigQuery data, proposes SQL fixes, and retrieves prior incidents from ChromaDB memory.

Ambiguity-Clarifying LLM | *Qwen 2.5, PyTorch, SFT, DPO, Hugging Face* GitHub
- Built a three-stage SFT/DPO pipeline for Qwen 2.5 and an LLM-as-a-judge harness over six quality dimensions, improving context-aware clarification accuracy by **25%**.

Contract Pricing Intelligence | *Django, REST APIs, PostgreSQL, AWS S3, Pandas* GitHub
- Shipped a full-stack Django/DRF prototype that parses contracts into pricing fields, exposes CRUD/REST workflows, validates rules, calculates analytics, and stores documents in AWS S3.

Real-Time Data Quality Pipeline | *Redpanda, Snowflake, dbt, Streamlit, Docker* GitHub
- Built a Dockerized synthetic-event pipeline from Redpanda to Snowflake with dbt freshness, duplicate, null, and integrity checks plus live Streamlit metrics and alerts.

Technical Skills

Languages: Python, SQL, R, Bash, JavaScript, HTML/CSS
Frameworks & Tools: PyTorch, TensorFlow, scikit-learn, XGBoost, LangChain, Hugging Face, Django/DRF, Node.js, Streamlit, Docker, Git, Claude Code, Cursor, Copilot
AI & Data Systems: Agentic workflows, RAG, SFT/DPO, LLM evaluation, PySpark, Airflow, dbt, Snowflake, BigQuery, AWS, PostgreSQL, MongoDB, **Neo4j**, ChromaDB, Redpanda, Docker